Welcome

COURSE:
ADVANCED PROJECT MANAGEMENT IN AI

Lesson 2 Session 1

Planning and Scheduling AI Projects:

AI Project Lifecycle and Stages

Welcome

课程：
人工智能高级项目管理

第2课 第1节

人工智能项目规划与进度安排：

人工智能项目生命周期与阶段

LESSON 2.1 – AI Project Lifecycle and Stages



第2.1课 - 人工智能项目生命周期与阶段



2. Learning Objectives: Lesson 2.1

- Explain the key stages of the AI project lifecycle
- Identify major lifecycle frameworks used in AI projects (CRISP-DM, TDSP, MLOps)
- Describe the activities and decisions involved at each stage of an AI project
- Recognize the project management implications of managing AI systems across their lifecycle

2. 学习目标：第2.1课

- 阐释人工智能项目生命周期的关键阶段
- 识别人工智能项目中使用的生命周期主要框架（CRISP-DM、TDSP、MLOps）
- 描述人工智能项目每个阶段涉及的活动与决策
- 认识到在整个生命周期中管理人工智能系统对项目管理的影响

Structuring an AI Project

AI projects require a structured lifecycle to manage:

- Multiple technical stages
- Cross-functional collaboration
- Iterative experimentation
- Integration with operational systems

Common lifecycle frameworks used in practice:

- CRISP-DM
- Team Data Science Process (TDSP)
- MLOps lifecycle models

These frameworks help translate business problems into deployable AI systems.

构建人工智能项目

人工智能项目需要结构化的生命周期来管理：

- 多个技术阶段
- 跨职能协作
- 迭代式实验
- 与运营系统的集成

实践中常用的生命周期框架：

- CRISP-DM（跨行业数据挖掘标准流程）
- TDSP（团队数据科学过程）
- MLOps（机器学习运营）生命周期模型

这些框架有助于将业务问题转化为可部署的人工智能系统。

The AI Project Lifecycle Overview

Typical AI lifecycle stages:

1. Problem definition
2. Data acquisition
3. Data preparation
4. Model development
5. Model validation
6. Deployment
7. Monitoring
8. Retraining

Lifecycle frameworks differ in terminology but follow similar logical stages. Project managers coordinate activities across all stages rather than focusing only on model development.

人工智能项目生命周期概述

典型的人工智能生命周期阶段包括：

1. 问题定义
2. 数据获取
3. 数据准备
4. 模型开发
5. 模型验证
6. 部署
7. 监控
8. 重新训练

生命周期框架在术语上有所差异，但遵循相似的逻辑阶段。
项目经理需协调所有阶段的活动，而不仅仅关注模型开发。

CRISP-DM Framework

CRISP-DM (Cross-Industry Standard Process for Data Mining)

Six stages:

1. Business Understanding
2. Data Understanding
3. Data Preparation
4. Modeling
5. Evaluation
6. Deployment



Characteristics:

- Iterative workflow
- Strong link between business goals and analytics
- Widely adopted across industries

CRISP-DM 框架

CRISP-DM（跨行业数据挖掘标准流程）

六个阶段：

1. 商业理解
2. 数据理解
3. 数据准备
4. 建模
5. 评估
6. 部署



特点：

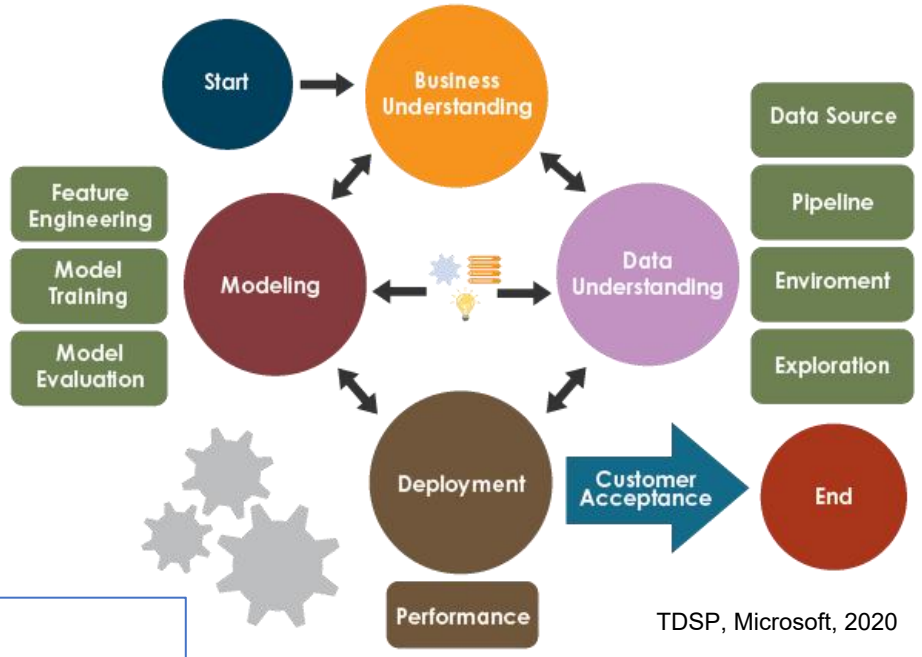
- 迭代式工作流
- 业务目标与分析之间的紧密联系
- 在跨行业中广泛采用

Enterprise Lifecycle: Team Data Science Process

Microsoft Team Data Science Process (TDSP)

Stages include:

- Business understanding
- Data acquisition and exploration
- Modeling
- Deployment
- Customer acceptance



TDSP, Microsoft, 2020

Key features:

- Team collaboration
- Standardized documentation
- Scalable enterprise workflows

企业生命周期：团队数据科学过程

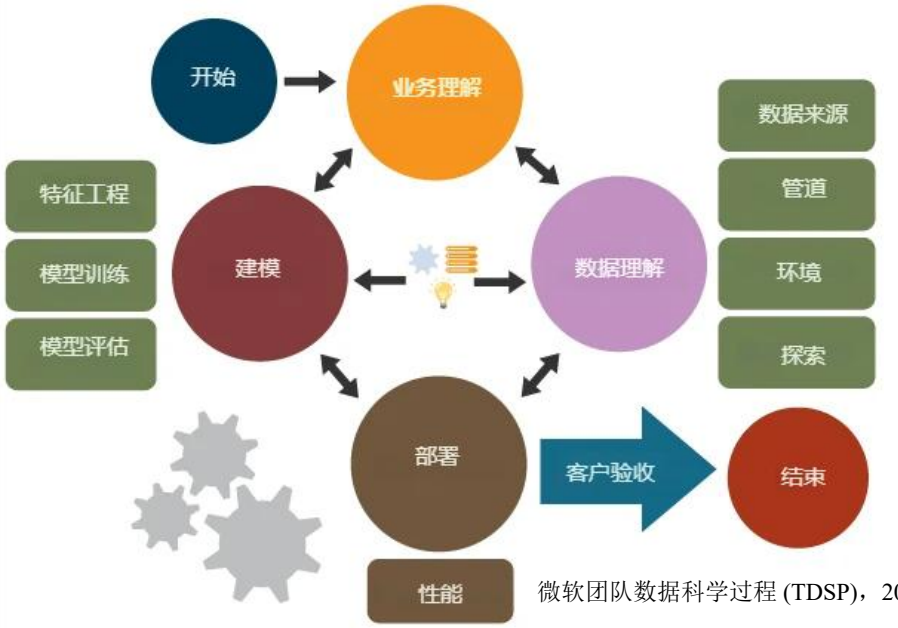
微软团队数据科学过程 (TDSP)

阶段包括：

- 商业理解
- 数据获取与探索
- 建模
- 部署
- 客户验收

主要特征：

- 团队协作
- 标准化文档
- 可扩展的企业工作流



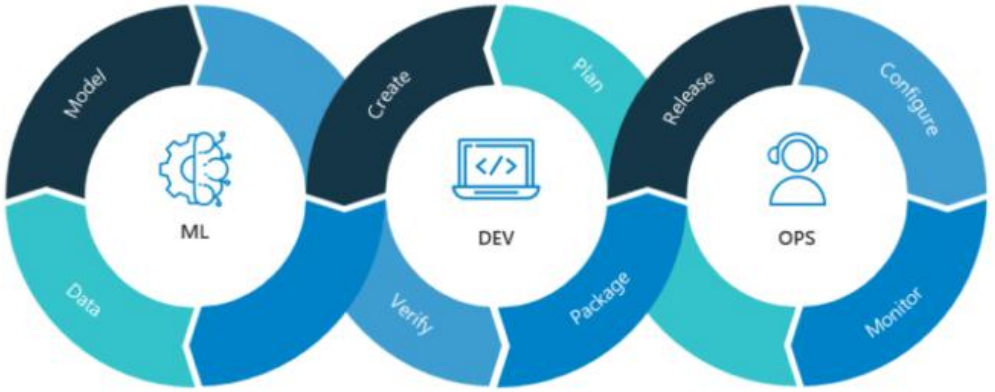
微软团队数据科学过程 (TDSP)，2020

MLOps Lifecycle

MLOps extends the lifecycle beyond development.

Core stages:

Data - Training - Validation - Deployment - Monitoring - Retraining



Nvidia, 2020

Purpose:

- Continuous model improvement
  - Automation of deployment pipelines
  - Operational reliability

MLOps connects machine learning with DevOps practices.

MLOps 生命周期

MLOps 将生命周期延伸至开发阶段之后。

核心阶段：

数据 → 训练 → 验证 → 部署 → 监控 → 重新训练



Nvidia, 2020

目标：

- 持续改进模型
- 自动化部署流水线
- 运营可靠性

MLOps 将机器学习与 DevOps 实践相结合。

Stage 1: Problem Definition

Key activities:

- Identify business objective
- Define ML prediction or classification task
- Establish measurable success criteria
- Assess feasibility of AI solution

Example:

Fraud detection - classify transactions as fraudulent or legitimate.

阶段 1：问题定义

关键活动：

- 明确业务目标
- 定义机器学习预测或分类任务
- 建立可衡量的成功标准
- 评估人工智能解决方案的可行性

示例：

欺诈检测——将交易分类为欺诈性或合法性交易。

Stage 2: Data Acquisition

Data acquisition considerations:

- Data availability
- Data ownership
- Privacy and regulatory compliance
- Data labeling requirements

Data feasibility often determines project viability.

阶段 2：数据获取

数据获取需考虑的因素：

- 数据可用性
- 数据所有权
- 隐私与合规性要求
- 数据标注需求

数据的可行性往往决定了项目的可行性。

Stage 3: Data Preparation

Typical activities:

- Cleaning missing or inconsistent data
- Removing duplicates
- Feature engineering
- Data transformation

Data preparation can account for 60–80% of project effort.

阶段 3：数据准备

典型活动包括：

- 清理缺失或不一致的数据
- 移除重复数据
- 特征工程
- 数据转换

数据准备可能占项目工作量的 60%–80%。

Stage 4: Model Development

Activities include:

- Algorithm selection
- Model training
- Hyperparameter tuning
- Feature selection

Multiple models are usually tested before selecting the final model.

阶段 4：模型开发

活动包括：

- 算法选择
- 模型训练
- 超参数调优
- 特征选择

在选定最终模型之前，通常会测试多个模型。

Stage 5: Model Validation

Validation techniques:

- Train-test split
- Cross-validation
- Precision and recall analysis
- Confusion matrices

Evaluation metrics must align with business objectives.

阶段 5：模型验证

验证技术：

- 训练-测试集划分
- 交叉验证
- 精确率与召回率分析
- 混淆矩阵

评估指标必须与业务目标保持一致。

Stage 6: Deployment

Deployment integrates the model into operational systems.

Examples:

- Payment fraud detection
- E-commerce recommendation engines
- Medical diagnostic tools

Deployment challenges:

- System integration
- Latency requirements
- Security constraints

阶段 6：部署

部署环节将模型集成到运营系统中。

示例：

- 支付欺诈检测
- 电子商务推荐引擎
- 医疗诊断工具

部署面临的挑战：

- 系统集成
- 延迟要求
- 安全约束

Stage 7: Monitoring and Model Drift

AI models degrade when data patterns change.

Causes include:

- Market changes
- Behavioral shifts
- Economic conditions

Monitoring metrics:

- Accuracy trends
- Data distribution changes
- System reliability

阶段 7：监控与模型漂移

当数据模式发生变化时，人工智能模型会性能下降。

原因包括：

- 市场变化
- 行为转变
- 经济状况

监控指标：

- 准确率趋势
- 数据分布变化
- 系统可靠性

Stage 8: Model Retraining

Retraining process:

- Update datasets
- Re-train models
- Re-validate performance
- Deploy updated versions

Many organizations implement automated retraining pipelines.

阶段 8：模型重新训练

重新训练流程：

- 更新数据集
- 重新训练模型
- 重新验证性能
- 部署更新后的版本

许多组织会实施自动化重新训练流水线。

Key Takeaways

AI lifecycle management requires:

- Coordination across multiple technical roles
- Iterative experimentation planning
- Data governance oversight
- Continuous system monitoring

Project managers oversee the entire lifecycle rather than individual stages.

核心要点

人工智能生命周期管理需要：

- 协调多个技术角色
- 规划迭代式实验
- 监督数据治理
- 持续监控系统

项目经理需统筹整个生命周期，而非仅关注单个阶段。

THANKS !!!



谢谢！



Welcome

COURSE:
ADVANCED PROJECT MANAGEMENT IN AI

Lesson 2 Session 2

Planning and Scheduling AI Projects:
Scope definition
and
Requirements Engineering

Welcome

课程：
人工智能高级项目管理

第2课 第2节

人工智能项目规划与进度安排：

范围定义

与

需求工程

LESSON 2.2 – Scope Definition and Requirements Engineering



第2.2课 – 范围定义与需求工程



2. Learning Objectives: Lesson 2.2

- Understanding how business problems map to machine learning objectives.
- Differentiating functional versus non-functional requirements.
- Specifying data and model requirements clearly.
- Managing evolving requirements and scope changes.

2. 学习目标：第2.2课

- 理解业务问题如何映射为机器学习目标
- 区分功能性与非功能性需求
- 明确说明数据与模型需求
- 管理不断变化的需求与范围变更

Why Scope & Requirements Matter in AI

- Clear scope prevents misalignment between business and AI outcomes
- Reduces rework due to misinterpreted objectives
- Supports realistic timeline & budget planning
- Mitigates ethical, regulatory, and compliance risks

为何范围与需求在人工智能项目中至关重要

- 清晰的范围可防止业务目标与人工智能成果之间出现偏差
- 减少因目标误解而造成的返工
- 为切实可行的时间线与预算规划提供支持
- 降低伦理、监管与合规风险

Translating Business Problem to ML Objective

- Identify the business goal (profit, efficiency, compliance)
- Determine measurable AI output (classification, prediction, recommendation)
- Define success metrics (accuracy, ROI, adoption)
- Examples:
  - Predictive maintenance → minimize downtime
  - Customer churn prediction → improve retention

将业务问题转化为机器学习目标

- 明确业务目标（利润、效率、合规性）
- 确定可衡量的人工智能产出（分类、预测、推荐）
- 定义成功标准（准确率、投资回报率、采纳率）
- 示例：
  - 预测性维护 ⟹ 最大限度减少停机时间
  - 客户流失预测 ⟹ 提升客户留存率

Functional Requirements in AI

- Describe what the system should do
- Examples:
  - ML model must classify transactions as fraudulent/non-fraudulent
  - Chatbot must answer customer queries with 90% accuracy
- Focus on outputs, interactions, model capabilities

人工智能中的功能性需求

- 描述系统应执行的功能
- 示例：
  - 机器学习模型必须将交易分类为欺诈性/非欺诈性
  - 聊天机器人必须以90%的准确率回答客户问题
- 侧重于输出、交互、模型能力

Non-Functional Requirements in AI

- Describe how the system should perform
- Examples:
  - Latency: response time < 2 seconds
  - Scalability: support 10k concurrent users
  - Reliability, maintainability, compliance

人工智能中的非功能性需求

- 描述系统应如何执行
- 示例：
  - 延迟：响应时间小于2秒
  - 可扩展性：支持1万并发用户
  - 可靠性、可维护性、合规性

Data Requirements Specification

- Define data needed for model training & validation
- Consider:
  - Source, type, volume, and frequency of data
  - Data quality & completeness
  - Labeling & feature engineering needs
- Example:
- Predictive maintenance

  3 years of sensor readings per machine

数据需求规格说明

- 定义模型训练与验证所需的数据
- 考虑因素：
  - 数据的来源、类型、体量与频率
  - 数据质量与完整性
  - 标注与特征工程需求
- 示例：
- 预测性维护 ⟹

  每台机器3年的传感器读数

Work Breakdown Structure (WBS) for AI

- Breaking project into manageable components
- Major phases / deliverables:
  - Problem definition
  - Data acquisition & preprocessing
  - Model development & validation
  - Deployment & monitoring
- WBS helps clarify dependencies and assign resources

人工智能项目的工作分解结构（WBS）

- 将项目分解为可管理的组成部分
- 主要阶段/可交付成果：
  - 问题定义
  - 数据获取与预处理
  - 模型开发与验证
  - 部署与监控
- 工作分解结构有助于厘清依赖关系并分配资源

Managing Evolving Requirements

- AI projects are inherently experimental
- Requirements evolve as models and data insights mature
- Strategies:
  - Maintain a flexible scope baseline
  - Use iterative requirement reviews
- Include stakeholder checkpoints

管理不断变化的需求

- 人工智能项目本质上具有实验性质
- 随着模型和数据洞察的成熟，需求会不断演变
- 策略：
  - 保持灵活的范围基线
  - 采用迭代式需求评审
- 设置利益相关方检查点

Scope Management Approaches

Waterfall: fixed scope, linear phases; easier for compliance-heavy projects

Agile: flexible scope, incremental delivery; suits experimental AI work

Hybrid: core scope fixed, experimentation areas iterative; balances predictability & flexibility

Key takeaway: Choose methodology based on uncertainty & risk

范围管理方法

瀑布式：范围固定，阶段线性；适用于合规性要求较高的项目

敏捷式：范围灵活，增量交付；适合实验性人工智能工作

混合式：核心范围固定，实验领域迭代；兼顾可预测性与灵活性

核心要点：根据不确定性与风险选择相应方法

Handling Scope Creep

- Scope creep: uncontrolled changes to project scope
- Causes in AI projects: new data insights, stakeholder requests, evolving regulations
- Strategies:
  - Change control process; document, review, approve changes
  - Prioritize backlog items (Agile) or update requirement baseline (Waterfall)
- Communicate impact on timeline, budget, and performance metrics

处理范围蔓延

- 范围蔓延：项目范围不受控制地扩大
- 在人工智能项目中的成因：新的数据洞察、利益相关方的新需求、不断变化的法规
- 策略：
  - 建立变更控制流程：记录、评审、批准变更
  - 对待办事项列表进行优先级排序（敏捷式）或更新需求基线（瀑布式）
- 沟通变更对时间线、预算和绩效指标的影响

Stakeholder Alignment

- Identify stakeholders: business, technical, regulatory
- Clarify expectations vs feasible outcomes
- Techniques:
  - Workshops & interviews
  - Requirement traceability matrix
- Joint review sessions

利益相关方协调

- 识别利益相关方：业务方、技术方、监管方
- 明确期望与可行成果之间的界限
- 技术手段：
  - 研讨会与访谈
  - 需求可追溯性矩阵
- 联合评审会议

Tools & Artifacts

- Requirements documentation: Business Requirements Document (BRD), Systems Requirement Specification (SRS), Market Requirements Document (MRD)
- Data specification templates
- Use case diagrams & user stories for AI features
- Traceability from business objective

→ ML requirement

→ model output

工具与工件

- 需求文档：业务需求文档（BRD）、系统需求规格说明（SRS）、市场需求文档（MRD）
- 数据规格模板
- 针对人工智能功能的用例图与用户故事
- 从业务目标

→ 机器学习需求

→ 模型输出的可追溯性

Common Pitfalls

- Vague or conflicting objectives
- Ignoring non-functional requirements
- Underestimating data needs
- Late stakeholder involvement
- Example:

AI project fails due to insufficient labeled data; retraining delays

常见陷阱

- 目标模糊或相互冲突
- 忽视非功能性需求
- 低估数据需求
- 利益相关方介入过晚
- 示例：

人工智能项目因标注数据不足而失败，导致重新训练延期

Summary

- Scope & requirements form the foundation of AI success
- Functional, non-functional, and data requirements are critical
- Flexibility and stakeholder alignment reduce risk
- Methodology choice affects how scope is managed
- Proactive handling of scope creep ensures project stability

总结

- 范围与需求构成人工智能成功的基石
- 功能性需求、非功能性需求与数据需求至关重要
- 灵活性与利益相关方协调可降低风险
- 方法论的选择影响范围管理的方式
- 主动应对范围蔓延可确保项目稳定

THANKS !!!



谢谢！



Welcome

COURSE:
ADVANCED PROJECT MANAGEMENT IN AI

Lesson 2 Session 3

Planning and Scheduling AI Projects:
Scheduling and Timeline Development
in AI Projects

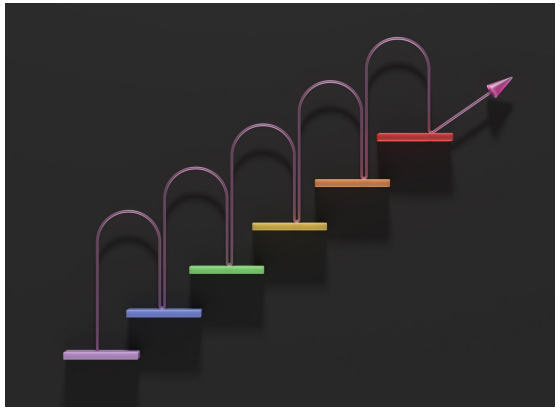
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课程：
人工智能高级项目管理

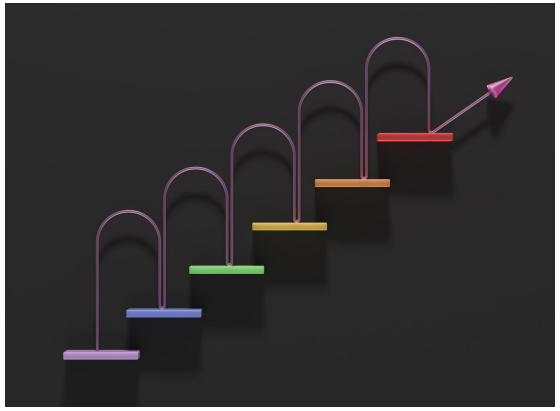
第2课 第3节

人工智能项目规划与进度安排：
人工智能项目中的进度安排
与时间线制定

LESSON 2.3– Scheduling and Timeline Development



第2.3课 – 进度安排与时间线制定



2. Learning Objectives: Lesson 2.3

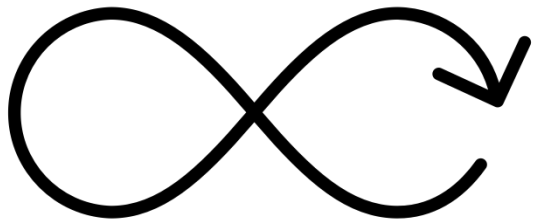
- Recognize unique scheduling challenges in AI projects
- Apply iterative milestone and hybrid planning approaches
- Utilize time management tools effectively
- Communicate uncertainty and buffer periods to stakeholders

2. 学习目标：第2.3课

- 识别人工智能项目中独特的进度安排挑战
- 应用迭代式里程碑与混合规划方法
- 有效运用时间管理工具
- 向利益相关方沟通不确定性及缓冲期

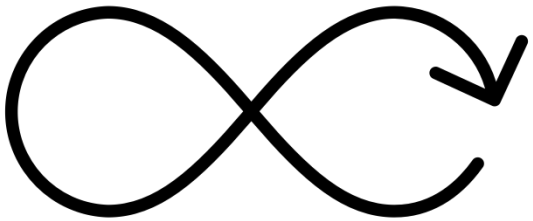
Why AI Project Scheduling is Unique

- High uncertainty due to data variability and model experimentation
- Traditional deterministic timelines often fail
- Iterative retraining cycles require flexible scheduling



为何人工智能项目进度安排具有独特性

- 数据可变性与模型实验带来了高度不确定性
- 传统的确定性时间线往往难以奏效
- 迭代式重新训练周期需要灵活的进度安排



Iterative Milestone Planning

- Milestones are non-linear and adaptive
- Examples: prototype completion, validation results, pilot deployment
- Early milestones help identify blockers and adjust schedules dynamically



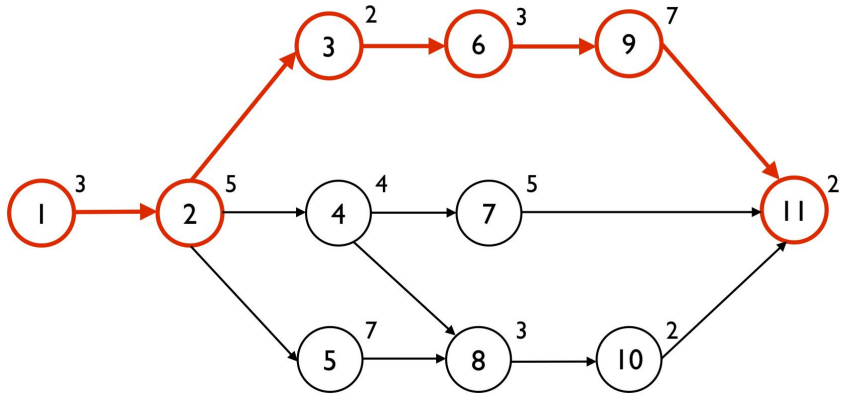
迭代式里程碑规划

- 里程碑是非线性的且具有适应性的
- 示例：原型完成、验证结果、试点部署
- 早期里程碑有助于识别阻碍因素并动态调整进度



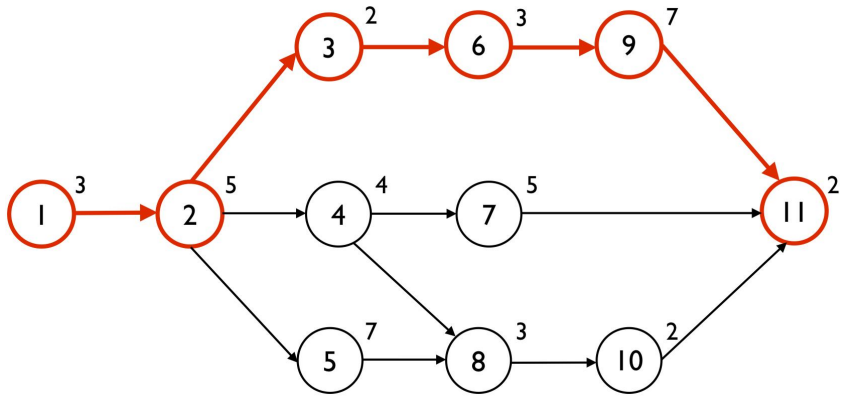
Critical Path Method (CPM)

- Focuses on tasks that directly affect project completion
- Highlights high-risk tasks prone to delays
- Incorporates model training and testing cycles into planning



关键路径法（CPM）

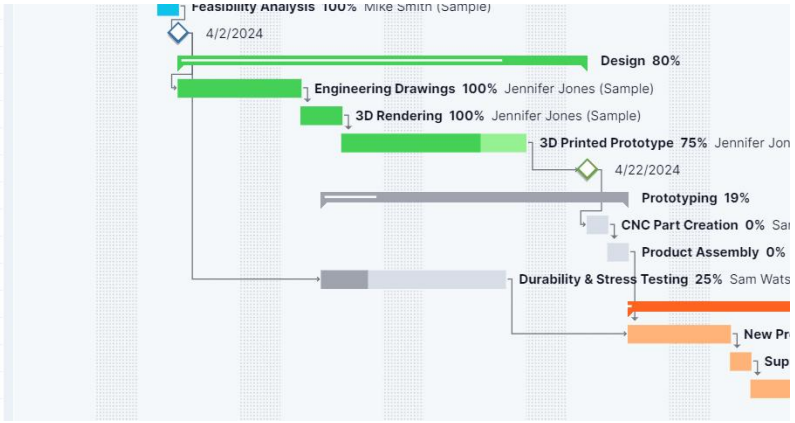
- 聚焦于直接影响项目完成的任务
- 突出容易发生延误的高风险任务
- 将模型训练与测试周期纳入规划



Gantt Charts for AI Projects

- Visualize tasks, durations, and dependencies
- Include buffer time for unexpected delays
- Helps align cross-functional teams
- Supports tracking iterative AI experiments

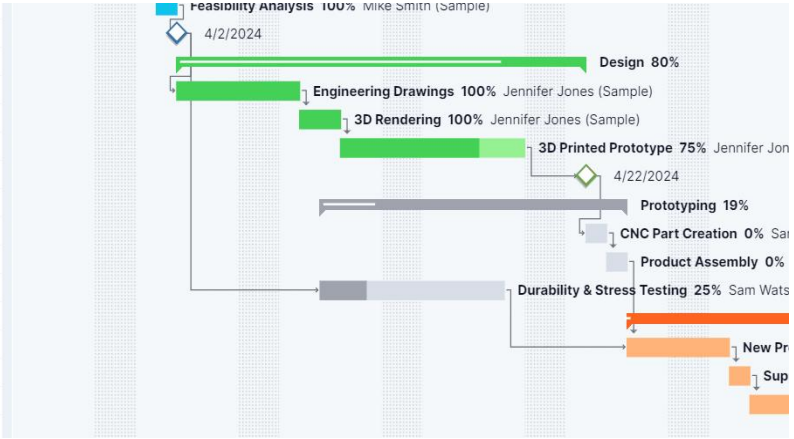
Feasibility Analysis	1.3	4/2/2024	1 day	$45.00
Stakeholder Feedback	1.4	4/2/2024	1 day	$720.00
Design	**2**	**4/3/2024**	**14 days**	**$2,560.00**
Engineering Drawings	2.1	4/3/2024	4 days	$640.00
3D Rendering	2.2	4/9/2024	2 days	$320.00
3D Printed Prototype	2.3	4/11/2024	7 days	$960.00
Stakeholder Feedback	2.4	4/22/2024	1 day	$640.00
Prototyping	**3**	**4/10/2024**	**11 days**	**$1,120.00**
CNC Part Creation	3.1	4/23/2024	1 day	$280.00
Product Assembly	3.2	4/24/2024	1 day	$280.00
Durability & Stress Te...	3.3	4/10/2024	7 days	$560.00
Pre-production	**4**	**4/25/2024**	**96 days**	
New Production Equip...	4.1	4/25/2024	3 days	
Supply Chain Sourcing	4.2	4/30/2024	1 day	
Product Documentation	4.3	5/1/2024	90 days	
Translation Services	4.4	9/4/2024	1 day	



适用于人工智能项目的甘特图

- 可视化展示任务、工期与依赖关系
- 为意外延误预留缓冲时间
- 有助于协调跨职能团队
- 支持追踪迭代式人工智能实验

Feasibility Analysis	1.3	4/2/2024	1 day	$45.00
Stakeholder Feedback	1.4	4/2/2024	1 day	$720.00
设计	**2**	**4/3/2024**	**14 days**	**$2,560.00**
Engineering Drawings	2.1	4/3/2024	4 days	$640.00
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Supply Chain Sourcing	4.2	4/30/2024	1 day	
Product Documentation	4.3	5/1/2024	90 days	
Translation Services	4.4	9/4/2024	1 day	



Accounting for Experimental Uncertainty

- AI project timelines are probabilistic, not deterministic
- Consider:
  - Data issues
  - Model convergence challenges
  - Regulatory reviews
- Build time buffers at critical phases

应对实验中的不确定性

- 人工智能项目的时间线是概率性的，而非确定性的
- 需考虑：
  - 数据问题
  - 模型收敛挑战
  - 监管评审
- 在关键阶段设置时间缓冲

Iteration and Feedback Loops

- Retraining and iteration are core to AI development
- Feedback loops from testing inform next phase adjustments
- Encourages continuous improvement without derailing schedules

迭代与反馈循环

- 重新训练与迭代是人工智能开发的核心
- 来自测试的反馈循环为下一阶段的调整提供依据
- 在保障进度不受影响的前提下，鼓励持续改进

Parallel vs Sequential Activities

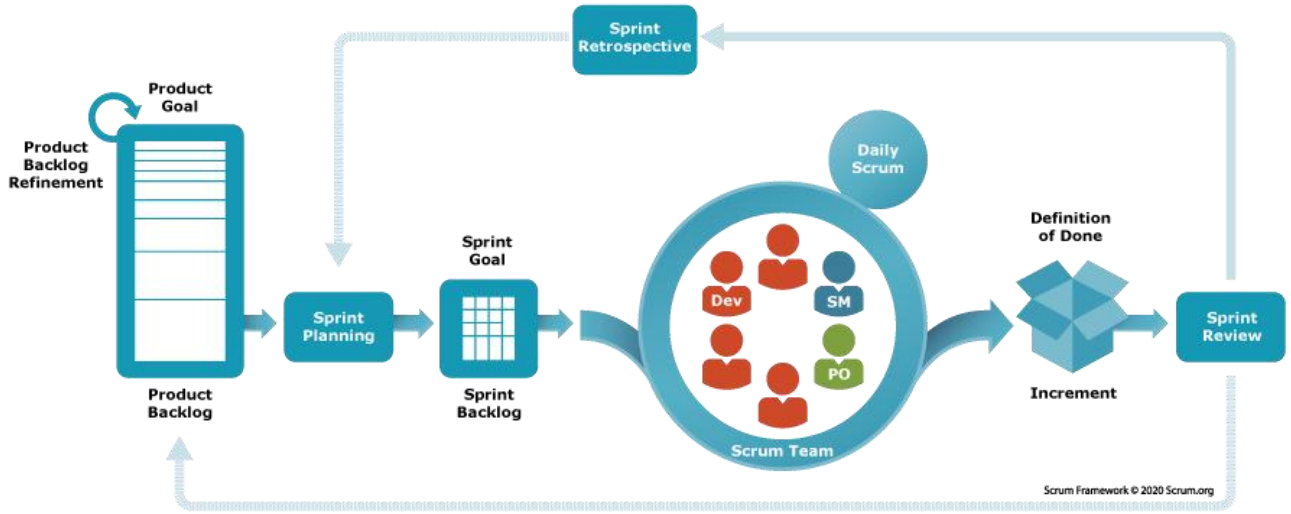
- Parallelizable tasks: data preprocessing, infrastructure setup, initial experiments
- Sequential dependencies: validation  deployment
- Effective parallelization reduces total project duration

并行活动与串行活动

- 可并行执行的任务：数据预处理、基础设施搭建、初步实验
- 存在串行依赖关系的任务：验证 → 部署
- 有效的并行化可缩短项目总工期

Agile/Hybrid Approaches

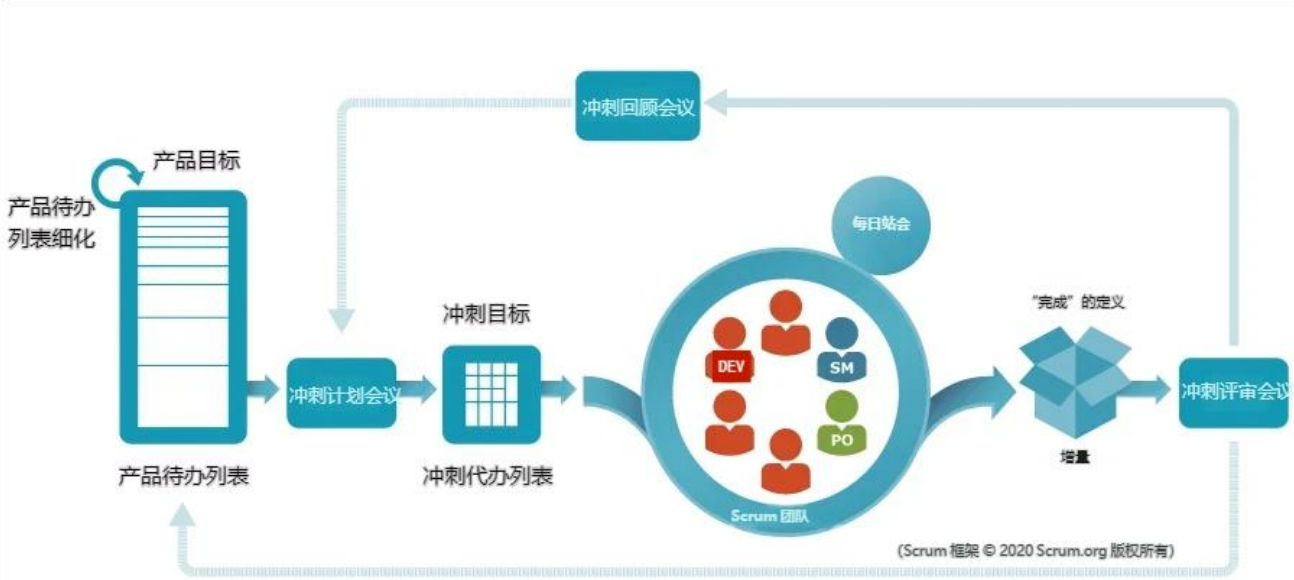
- Iterative sprints applied to model development cycles
- Hybrid planning combines predictive and adaptive elements
- Balances business deadlines with experimental uncertainty



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敏捷/混合方法

- 将迭代式冲刺应用于模型开发周期
- 混合规划结合了预测性要素与适应性要素
- 在业务截止日期与实验不确定性之间取得平衡



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Stakeholder Communication

- Present timeline uncertainty and buffers clearly
- Align expectations with iteration-dependent milestones
- Builds trust and realistic understanding of AI project timelines

利益相关方沟通

- 清晰说明时间线的不确定性及缓冲设置
- 使期望与依赖迭代的里程碑保持一致
- 建立对人工智能项目时间线的信任与切合实际的理解

Tools and Techniques

- Software: Microsoft Project, Jira, Smartsheet, Asana
- Techniques: CPM, Gantt charts, milestone-based iterative planning
- Integration with MLOps pipelines for progress tracking

工具与技术

- 软件工具：Microsoft Project、Jira、Smartsheet、Asana
- 技术方法：关键路径法、甘特图、基于里程碑的迭代式规划
- 与MLOps流水线集成以跟踪进度

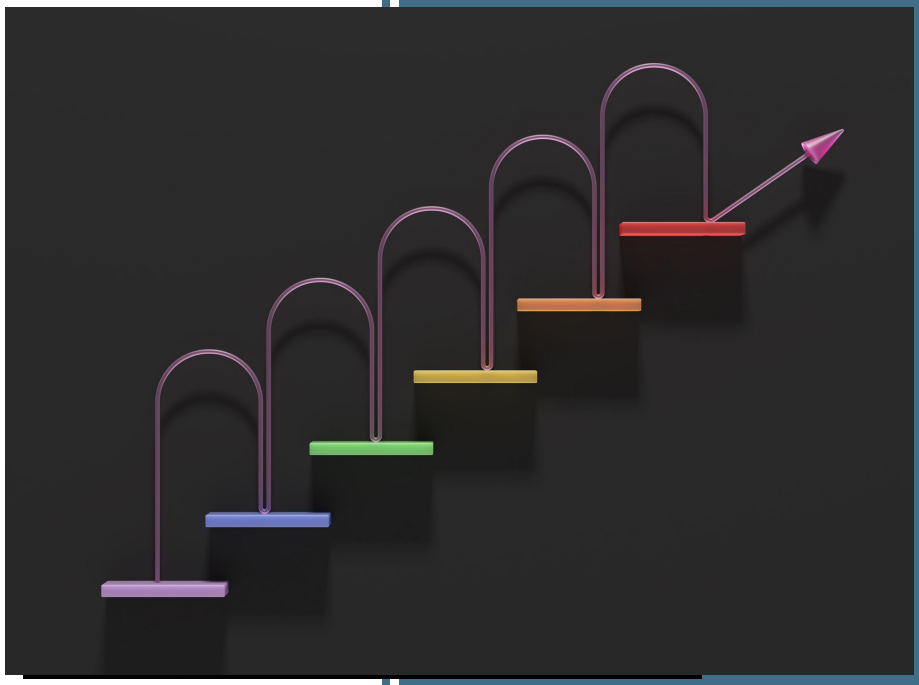
Key Takeaways

- AI project schedules require flexibility and iterative planning
- Buffers, feedback loops, and parallelization mitigate risk
- Transparent communication ensures stakeholder alignment

核心要点

- 人工智能项目进度安排需要灵活性与迭代式规划
- 缓冲设置、反馈循环与并行化可降低风险
- 透明的沟通可确保利益相关方保持一致

THANKS !!!



谢谢！

